

EPI Distortion Correction without Opposite Phase Encodings with Unsupervised INR-Based Deformable Registration

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Journal club presentation

- **Why is EPI distortion correction problem important?**

EPI distortion

Susceptibility field inhomogeneity causes geometric warping, mainly along the phase-encoding direction. Severe errors can reach tens of millimeters in frontal, temporal, and brainstem regions.

Topup / reverse PE

Reverse-PE methods can be very accurate, but they require an auxiliary b0 image acquired with the opposite phase encoding direction.

Practical gap

Many older or public dMRI datasets lack the reverse-PE acquisition. Classical T1-based registration methods are more widely applicable but less accurate.

EPINR question

Can a continuous coordinate-based neural field recover a physically PE displacement field from only a distorted b0 and T1w image?

Core challenge: recover a distortion field without the usual reverse-PE information.

- **Main idea**

EPINR learns a smooth displacement field for one subject by warping a distorted single-PE b0 image toward a T1-weighted anatomical reference.

Input	Mechanism	Output
Distorted b0 + T1w (no reverse PE)	Coordinate-based SIREN models PE displacement	Corrected b0 + estimated displacement field

- No auxiliary opposite phase-encoding image is required
- Model is optimized separately for every subject
- Validated on two diffusion MRI datasets with different image characteristics

- **What is an implicit neural representation?**

Conventional voxel field

A displacement map is stored explicitly as voxel values:

$VDM[i,j,k]$

INR field

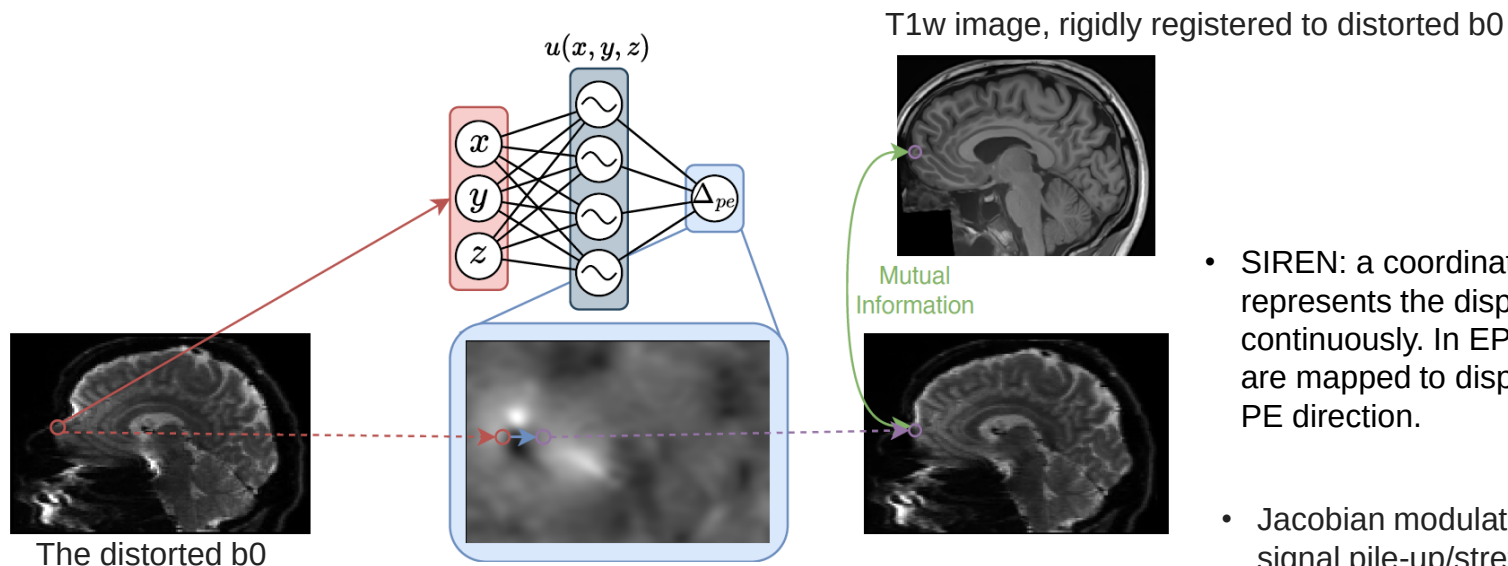
A neural function maps coordinates to values:

$u_{\theta}(x,y,z) \rightarrow \Delta y$

In EPINR, the network does not directly take a whole image

It is queried at spatial coordinates and returns a smooth displacement along the PE direction.

• EPINR model overview



- SIREN: a coordinate-based MLP that represents the displacement field continuously. In EPINR, 3D coordinates are mapped to displacement along the PE direction.
- Jacobian modulation accounts for signal pile-up/stretching

Figure 1: Illustration of our proposed EPINR registration model. EPINR learns to warp a distorted b0 to a non-distorted T1w reference in an unsupervised manner with a displacement field (blue) constrained in the phase encoding direction. The C^∞ smoothness of EPINR also allows for *analytically* calculating higher-order regularization methods and Jacobian modulation (Jezzard and Balaban, 1995).

• Loss function

EPINR is trained without supervised displacement labels

$$\mathcal{L}_{\text{sim}} = \frac{1}{N} \sum_{i=1}^N -\text{MI}(\det J_u(\mathbf{c}_i + u_\theta(\mathbf{c}_i)) \mathbf{M}[\mathbf{c}_i + u_\theta(\mathbf{c}_i)], \mathbf{F}[\mathbf{c}_i]),$$

$$\mathcal{L}_{\text{lap}} = \frac{1}{N} \sum_{i=1}^N -\text{NCC}(\nabla^2(\det J_u(\mathbf{c}_i + u_\theta(\mathbf{c}_i)) \mathbf{M}[\mathbf{c}_i + u_\theta(\mathbf{c}_i)]), \nabla^2(1 - \mathbf{F}[\mathbf{c}_i])),$$

$$\mathcal{L}_{\text{smooth}} = \|J_u(\mathbf{c}_i + u_\theta(\mathbf{c}_i))\|_F = \left| 1 + \frac{\partial u_\theta(\mathbf{c}_i)}{\partial y} \right|$$

$$\mathcal{L}_{\text{bend}} = \sum_{p=1}^3 \sum_{q=1}^3 \sum_{r=1}^3 \left(\mathbf{H}_p^{(u)}(\mathbf{c}_i + u_\theta(\mathbf{c}_i)) \right)_{q,r}^2 = \sum_{q=1}^3 \sum_{r=1}^3 \left(\mathbf{H}_{p=2}^{(u)}(\mathbf{c}_i + u_\theta(\mathbf{c}_i)) \right)_{q,r}^2,$$

$$\underbrace{\mathcal{L}_{\text{sim}} + \mathcal{L}_{\text{lap}}}_{\text{make corrected } b_0 \text{ agree with T1}} + \underbrace{\mathcal{L}_{\text{smooth}} + \mathcal{L}_{\text{bend}}}_{\text{keep deformation physically reasonable}}$$

1. Mutual information

Align corrected b0 with T1w despite different image contrasts.

2. Laplacian similarity

Encourages anatomical edge consistency after intensity inversion of T1w.

3. Smoothness

Regularizes local deformation using the computed Jacobian

4. Bending energy

Penalizes second-order curvature using analytical Hessian.

- **Data and preprocessing**

MICA-MICS

N = 49 subjects
AP b0 evaluated
1.6 mm isotropic dMRI
T1w: 0.8 mm isotropic

VCU MS

N = 48 subjects
10 controls + 38 MS patients
1.875×1.875×2 mm dMRI
T1w: 1 mm isotropic

Preprocessing pipeline

- DWI and T1w denoising and bias correction
- Rigid T1w-to-distorted-b0 registration with ANTs
- Creating masks using BET for b0 and SynthStrip for T1
- Topup with AP/PA b0s used as the reference for evaluation

- **Comparison methods**

Uncorrected

Preprocessed b0 with no susceptibility distortion correction.

ANTs-SyN

General deformable registration.

QSIPrep

Fieldmapless SDC in the QSIPrep pipeline

Synb0

GAN synthesizes an undistorted b0; topup uses real + synthetic b0 for correction.

EPINR

Subject-specific INR optimized from distorted b0 and T1w; no reverse PE needed.

Topup GT

Reverse-PE topup result used as silver-standard reference evaluation.

Evaluation metrics

- b0-topup MSE and susceptibility-field-topup MSE
- MI and LNCC between corrected b0 and T1w

• Quantitative results

Dataset	Model	b0-topup MSE $\times 10^{-6}$ ↓	SF-topup MSE $\times 10^{-3}$ ↓	MI b0-Anat. ↑	LNCC b0-Anat. ↑
MICA-MICS N=49	Uncorrected	1.3021 (0.300)	0.5397 (0.125)	0.3675 (0.034)	0.2271 (0.014)
	ANTs-SyN	1.6665 (0.354)	0.5229 (0.092)	0.3342 (0.054)	0.2395 (0.017)
	QSIRep	1.4983 (0.361)	0.5710 (0.128)	0.3677 (0.033)	0.2252 (0.012)
	Synb0+Topup	1.3741 (0.338)	1.4491 (0.267)	0.4285 (0.034)	0.2138 (0.013)
	EPINR	0.9878 (0.237)	0.4221 (0.087)	0.4299 (0.051)	0.2501 (0.016)
	Topup GT	-	-	0.4629 (0.039)	0.2639 (0.019)
VCU MS N=48	Uncorrected	4.1649 (1.375)	0.8881 (0.248)	0.2750 (0.040)	0.1621 (0.016)
	ANTs-SyN	3.5424 (1.448)	0.7522 (0.189)	0.2167 (0.039)	0.1840 (0.016)
	QSIRep	3.2523 (1.066)	0.8075 (0.222)	0.2667 (0.038)	0.1720 (0.017)
	Synb0+Topup	3.3660 (1.324)	2.7373 (0.900)	0.3276 (0.023)	0.1540 (0.016)
	EPINR	2.2992 (1.798)	0.6569 (0.526)	0.3690 (0.031)	0.1877 (0.018)
	Topup GT	-	-	0.3146 (0.027)	0.1780 (0.020)

Table 1: Results of different EPI distortion correction methods applied to two DWI datasets. Metric name arrows indicate the direction of better performance, and the best performance between all models is bolded.

- EPINR gives the lowest b0-topup and susceptibility-field MSE on both datasets
- It also achieves the highest b0-anatomy comparison

• Qualitative results

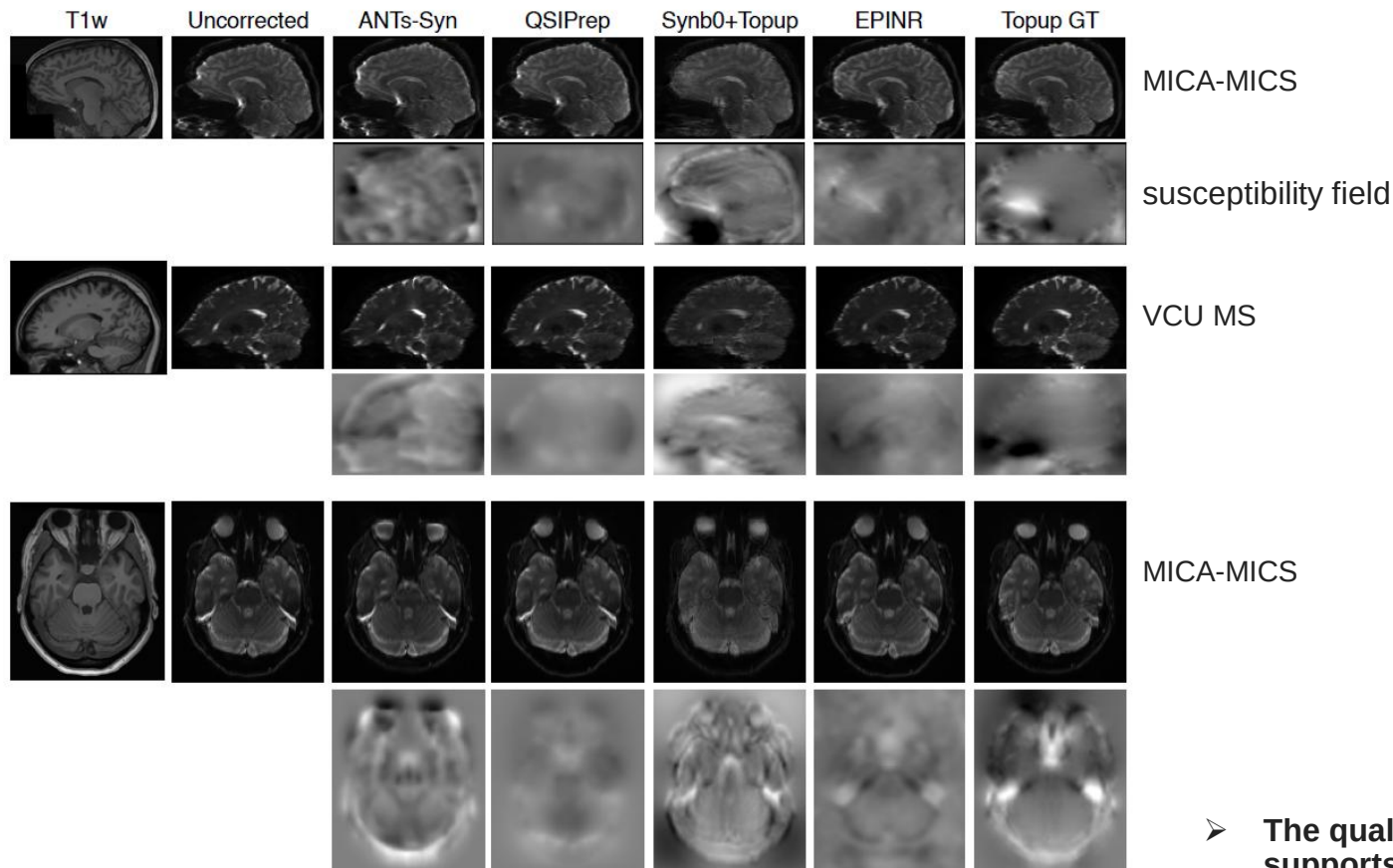


Figure 2: Qualitative results of susceptibility distortion correction over three subjects. Below each undistorted b0, the predicted susceptibility field in Hz is displayed. From top to bottom, subjects are from MICA-MICS, VCU MS, and MICA-MICS.

➤ The qualitative comparison supports the quantitative trend, but the authors caution that b0-T1 similarity alone can be misleading

• Ablation study

Dataset	Model	b0-topup MSE $\times 10^{-6} \downarrow$	SF-topup MSE $\times 10^{-3} \downarrow$	MI b0-Anat. $\uparrow$	LNCC b0-Anat. $\uparrow$
MICA-MICS N=10	Uncorrected	1.3488 (0.259)	0.5784 (0.135)	0.3849 (0.038)	0.2304 (0.018)
	No Regular.	1.8039 (0.399)	1.0106 (0.251)	0.4575 (0.077)	0.2539 (0.020)
	No Lapl. Sim.	1.0700 (0.246)	0.4942 (0.133)	<u>0.4500 (0.061)</u>	0.2514 (0.019)
	No Bend. Regular.	1.0214 (0.238)	0.4778 (0.124)	<u>0.4491 (0.058)</u>	0.2547 (0.020)
	No Jac. Mod.	0.9232 (0.156)	0.4315 (0.082)	0.4069 (0.040)	0.2519 (0.021)
	EPINR-128x3	<u>0.9577 (0.238)</u>	0.5112 (0.140)	0.4400 (0.055)	0.2494 (0.019)
	EPINR	<u>1.0054 (0.220)</u>	<u>0.4710 (0.124)</u>	0.4466 (0.063)	<u>0.2543 (0.020)</u>
	Topup GT	-	-	0.4835 (0.045)	0.2681 (0.021)
VCU MS N=10	Uncorrected	4.1100 (1.229)	0.8621 (0.298)	0.2680 (0.040)	0.1627 (0.013)
	No Regular.	2.8701 (1.582)	1.5217 (0.933)	0.3886 (0.040)	0.1860 (0.021)
	No Lapl. Sim.	2.6543 (1.200)	0.7439 (0.475)	0.3699 (0.036)	0.1853 (0.017)
	No Bend. Regular.	2.6021 (1.238)	0.7357 (0.529)	<u>0.3713 (0.036)</u>	0.1882 (0.017)
	No Jac. Mod.	3.2984 (1.424)	0.8411 (0.563)	<u>0.2655 (0.040)</u>	0.1821 (0.017)
	EPINR-128x3	2.6682 (1.249)	0.7803 (0.560)	0.3546 (0.032)	0.1847 (0.017)
	EPINR	<u>2.6046 (1.247)</u>	<u>0.7400 (0.540)</u>	0.3712 (0.037)	<u>0.1878 (0.017)</u>
	Topup GT	-	-	0.3057 (0.024)	0.1760 (0.013)

Table 2: Ablation results for EPINR models applied to 20 randomly selected subjects (10 from each dataset). Arrows indicate the direction of better performance, the best score is bolded, and the second best is underlined.

Ablation message: performance comes from the combination of smooth INR representation, physics-inspired constraints, and subject-specific optimization.

- **Interpretation and limitations**

Why EPINR works

A continuous coordinate-based neural field can recover a physically PE displacement field from only a distorted b0 and T1w image

Important cautions

- High b0-T1 similarity may reflect overfitting
- Corrected image similarity is not sufficient alone
- Subject-specific optimization is slower than direct inference

Runtime

Reported runtime: EPINR ~45–60 min per subject; ANTs-SyN ~1 min; Synb0+Topup ~7 min; QSIPrep ~60–90 min up to SDC.

Bottom line: EPINR improves correction quality, but speed and overfitting control are key practical issues.

Conclusion

- EPINR proposes an unsupervised INR-based registration method for EPI distortion correction without opposite PE acquisitions
- The method learns a smooth PE-constrained displacement field per subject from distorted b0 and T1w images
- Results show strong quantitative and qualitative performance against traditional and learning-based baselines
- Main limitations are runtime and the risk of overfitting to anatomical similarity metrics

Thank you